

Test of Mathematics for University Admission
Mock Set I Paper 1
 Applications of Mathematical Knowledge

Time allowed: **75 minutes**
 Questions: 20 multiple choice
 Calculators: not permitted
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Score: _____ / 20

Answer all 20 questions. Each question has exactly one correct option and is worth one mark. No marks are deducted for wrong answers. Every question uses AS content only. The difficulty lies in how that content is used: parameters that change the case, conditions that quietly remove a solution, and answers that come from structure rather than from brute force.

[illegible]

Question 1

The curve C has equation $x^2 + xy + y^2 = 3$. It is a closed curve, tilted relative to the axes. There are exactly two points on C at which the tangent to C is parallel to the line $y = x$. Find the distance between these two points.

- A $2\sqrt{3}$
- B $2\sqrt{6}$
- C 4
- D $\sqrt{6}$
- E $3\sqrt{2}$
- F $2\sqrt{2}$

Question 2

An arithmetic progression has first term a and common difference d , where a and d are both non-zero. Its 2nd, 5th and 11th terms, in that order, are three consecutive terms of a geometric progression. It turns out that the sum of the first ten terms of the arithmetic progression is itself one of the terms of the progression. Which term is it?

- A the 30th
- B the 32nd
- C the 55th
- D the 64th
- E the 65th
- F the 66th

Question 3

For which real values of a does the equation

$$|x - a| + |x - 2a| = a^2$$

have exactly two real solutions for x ?

- A $a \neq 0$
- B $|a| \geq 1$
- C $a > 1$
- D $|a| > 1$
- E $0 < |a| < 1$
- F $a < -1$ or $a > 2$

Question 4

For how many integers a does the curve $y = x^4 - 8x^2 + ax$ have exactly three stationary points?

- A 11
- B 23
- C 25
- D 27
- E 13
- F infinitely many

Question 5

For which real values of k does the equation

$$x^2 e^{-x} = k$$

have exactly three real solutions?

- A $0 < k < \frac{4}{e^2}$
- B $0 \leq k \leq \frac{4}{e^2}$
- C $0 < k < \frac{1}{e}$
- D $k > \frac{4}{e^2}$
- E $0 < k < \frac{2}{e}$
- F $k = \frac{4}{e^2}$ only

Question 6

The cubic $f(x) = x^3 + ax^2 + bx + c$ satisfies $f(0) = 0$ and $f(1) = 1$, and it is increasing: $f'(x) \geq 0$ for every real x . Find the set of all possible values of $f(2)$.

- A $f(2) = 8$ only
- B $0 \leq f(2) \leq 8$
- C $2 < f(2) < 8$
- D $f(2) \geq 2$
- E $2 \leq f(2) \leq 8$
- F $4 \leq f(2) \leq 8$

Question 7

For which values of a does the equation $x^3 - 3ax + 2 = 0$ have three distinct real roots, all greater than -3 ?

- A $1 < a < \frac{25}{9}$
- B $a > 1$
- C $0 < a < \frac{25}{9}$
- D $1 < a < 9$
- E $1 < a < 3$
- F no value of a

Question 8

In triangle ABC , $AB = x$, $AC = y$ and angle $BAC = 120^\circ$. The bisector of angle A meets BC at D , with $AD = 2$. Given also that $x + y = 12$, find BC .

- A $4\sqrt{6}$
- B $2\sqrt{33}$
- C $6\sqrt{3}$
- D $2\sqrt{30}$
- E $\sqrt{132}$
- F 8

Question 9

For $a > 0$ let

$$I(a) = \int_0^3 |x^2 - ax| dx.$$

Find the minimum value of $I(a)$ over all $a > 0$.

- A $\frac{9}{2}$
- B $\frac{9}{4}$
- C $9 - 3\sqrt{3}$
- D $\frac{9\sqrt{2}}{2}$
- E 3
- F $\frac{18 - 9\sqrt{2}}{2}$

Question 10

Let $k > 0$ be a constant. For each real number t , the tangent to the curve $y = x^2$ at the point (t, t^2) meets the curve $y = x^2 - k$ at two points, and the tangent and this second curve together enclose a finite region R_t . It turns out that the area of R_t is the same for every value of t . Given that this area is 36, find k .

- A 3
- B 6
- C 9
- D 27
- E $3\sqrt[3]{4}$
- F 12

Question 11

For which real values of k does the equation

$$3 \sin x + 4 \cos x = k$$

have exactly one solution with $0 \leq x \leq \pi$?

- A $-4 \leq k < 4$, or $k = 5$
- B $-5 \leq k \leq 5$
- C $-4 < k < 4$
- D $-4 \leq k \leq 4$
- E $k = -5$ or $k = 5$
- F $4 < k < 5$

Question 12

Let $m \geq 2$ be an integer. Four different numbers are chosen at random from the integers $1, 2, 3, \dots, 3m$. Which of the following is the probability that the sum of the four numbers is divisible by 3?

- A $\frac{1}{3} - \frac{1}{9m}$
- B $\frac{1}{3}$, whatever the value of m
- C $\frac{1}{3} + \frac{2}{9(3m-1)}$
- D $\frac{m}{3m-1}$
- E $\frac{m+1}{3m+1}$
- F it depends on m and has no simple form

Question 13

The curve with parametric equations

$$x = t^2 - 1, \quad y = t^3 - t \quad (t \in \mathbb{R})$$

crosses itself at exactly one point, forming a closed loop. Find the area enclosed by the loop.

- A $\frac{4}{15}$
- B $\frac{8}{15}$
- C $\frac{2}{3}$
- D $\frac{16}{15}$
- E $\frac{1}{2}$
- F $\frac{4}{5}$

Question 14

A square of side 1 is rotated about its centre through an acute angle θ . The region common to the original square and the rotated square has area $\frac{4}{5}$. Find the value of $\tan \theta + \frac{1}{\tan \theta}$.

- A $\frac{5}{4}$
- B $\frac{25}{12}$
- C $\frac{8}{5}$
- D $\frac{12}{5}$
- E 2
- F $\frac{16}{5}$

Question 15

Numbers a and b are chosen independently and uniformly at random from the interval $-1 \leq t \leq 1$. What is the probability that the equation

$$x^2 + 2ax + b = 0$$

has two distinct real roots, both strictly between -1 and 1 ?

- A $\frac{1}{3}$
- B $\frac{1}{4}$
- C $\frac{1}{12}$
- D $\frac{2}{3}$
- E $\frac{1}{6}$
- F $\frac{1}{8}$

Question 16

The curve $y = x^3 + ax^2 + bx$, where a and b are real constants, has two distinct stationary points. For which values of a and b does the straight line through these two stationary points pass through the origin?

- A exactly when $a = 0$
- B exactly when $b = 0$
- C exactly when $ab = 0$
- D exactly when $a^2 = 4b$
- E exactly when $a = b$
- F for every such a and b

Question 17

Evaluate

$$\int_0^{\pi/2} \frac{\sin^3 x}{\sin x + \cos x} dx.$$

- A $\frac{\pi - 1}{4}$
- B $\frac{\pi - 2}{4}$
- C $\frac{\pi}{8}$
- D $\frac{\pi + 1}{4}$
- E $\frac{\pi}{4} - \frac{1}{2}$
- F $\frac{1}{2}$

Question 18

Let $n \geq 2$ be an integer, and let $N(n)$ be the number of solutions of

$$\sin(nx) = \sin x \quad \text{with } 0 \leq x < 2\pi.$$

Which of the following is correct?

- A $N(n) = 2n - 2$ when n leaves remainder 1 on division by 4, and $N(n) = 2n$ otherwise
- B $N(n) = 2n$ for every n
- C $N(n) = 2n - 2$ when n is odd, and $N(n) = 2n$ when n is even
- D $N(n) = 2n - 1$ for every n
- E $N(n) = 2n + 2$ when n is even, and $N(n) = 2n$ when n is odd
- F $N(n) = 2n - 2$ when n is even, and $N(n) = 2n$ when n is odd

Question 19

As a and b range over all real numbers, find the minimum value of

$$\int_0^1 (x^2 - ax - b)^2 dx.$$

- A $\frac{1}{12}$
- B $\frac{1}{45}$
- C $\frac{1}{80}$
- D $\frac{1}{180}$
- E 0
- F $\frac{1}{120}$

Question 20

A point is chosen uniformly at random inside a square of side 1. What is the probability that it is closer to the nearest side of the square than to the centre of the square?

- A $\frac{\pi}{4}$
- B $\frac{8 - 4\sqrt{2}}{3}$
- C $\frac{3}{4}$
- D $\frac{4\sqrt{2} - 2}{3}$
- E $\frac{2}{3}$
- F $\sqrt{2} - \frac{1}{2}$

END OF PAPER